

CS57300
PURDUE UNIVERSITY

DATA MINING

ANNOUNCEMENT

- ▶ Expect assignment 3 to be release today
 - ▶ Due on October 20, 2023 (11:59pm)
 - ▶ Start early
- ▶ Midterm on October 18, 8-10pm
 - ▶ Location: BHEE 170
- ▶ If you have accessibility needs, please contact DRC to reserve a testing room with them

LOGISTIC REGRESSION: LEARNING

LOGISTIC REGRESSION

- ▶ Same parametric form as standard regression, but uses logistic function for binary classification

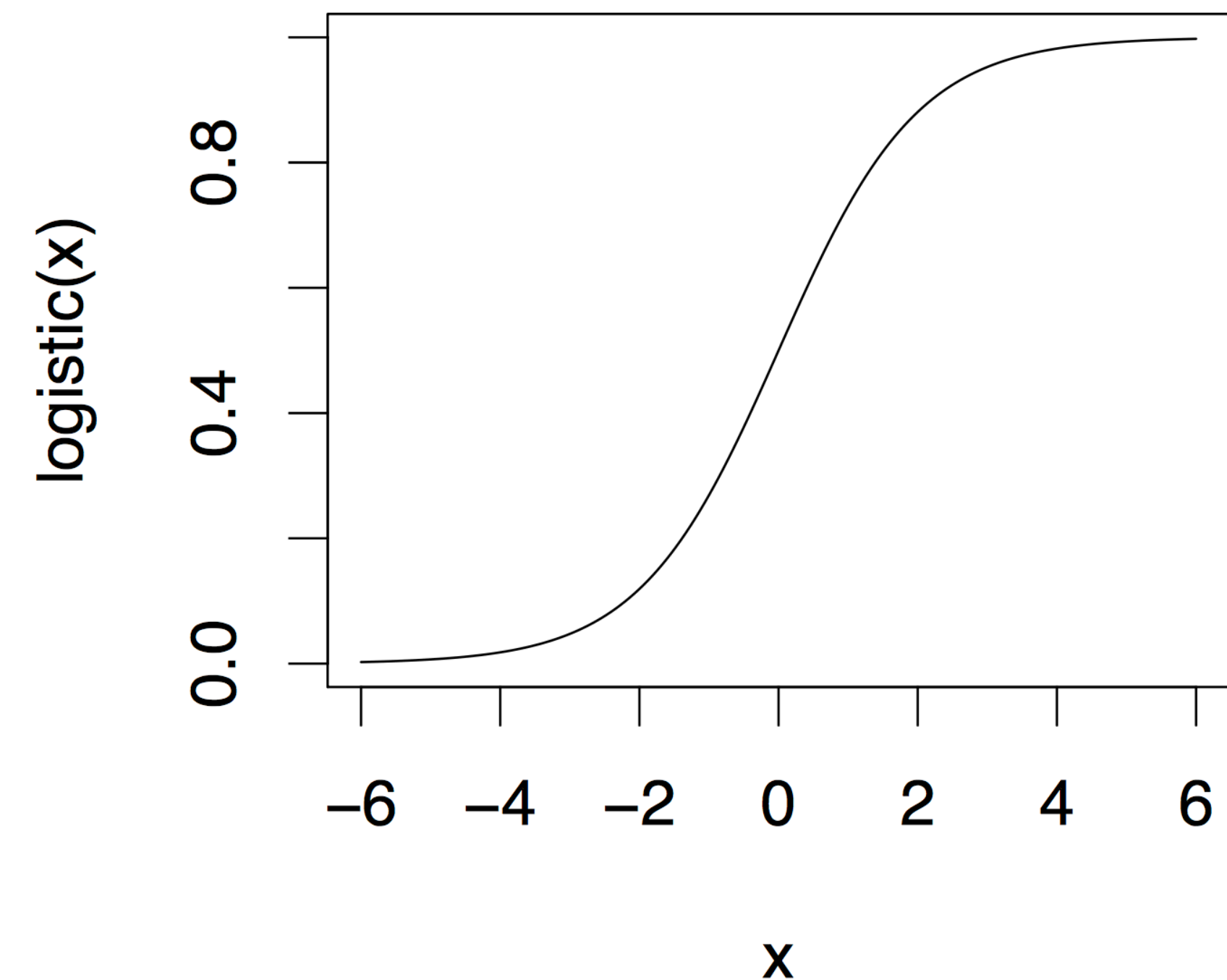
Logistic regression model:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + w_0)}}$$

- ▶ Output is the (positive) class probability rather than the binary prediction
- ▶ Logistic function transform ensures output is $[0,1]$

Logistic function:

$$\text{logistic}(x) := \frac{e^x}{1 + e^x} = \frac{1}{1 + e^{-x}}$$



LR EXAMPLE

$$P(BC = 1|A, I, S, CR) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x})}}$$

$$\mathbf{x} = [Int, A, I, S, CR]$$

$$\mathbf{w} = [w_0, w_A, w_I, w_S, w_{CR}]$$

LR parameters = $\mathbf{w}$

Intercept	Age>40	Income=high	Student=yes	Credit=fair	BuysComp?
1	0	1	0	1	0
1	0	1	0	0	0
1	0	1	0	1	1
1	1	0	0	1	1
1	1	0	1	1	1
1	1	0	1	0	0
1	0	0	1	0	1
1	0	0	0	1	0
1	0	0	1	1	1
1	1	0	1	1	1
1	0	0	1	0	1
1	0	0	0	0	1
1	0	1	1	1	1
1	1	0	0	0	0

- ▶ Score function: likelihood
- ▶ Estimate $\mathbf{w}$ with maximum likelihood estimation

LOGISTIC REGRESSION LEARNING

- ▶ Logistic regression: $P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + w_0)}}$
- ▶ Maximize (log) likelihood: $\mathbf{w} = (\mathbf{w}, w_0), \mathbf{x}_i = (\mathbf{x}_i, 1)$

$$\begin{aligned} \log L(\mathbf{w} | D) &= \sum_{i=1}^N \log p(y_i | \mathbf{w}) \\ &= \sum_{i=1}^N \log \left[\left(\frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}_i}} \right)^{y_i} \left(\frac{e^{-\mathbf{w}^T \mathbf{x}_i}}{1 + e^{-\mathbf{w}^T \mathbf{x}_i}} \right)^{1-y_i} \right] \\ &= \sum_{i=1}^N (y_i \mathbf{w}^T \mathbf{x}_i - \log(1 + e^{\mathbf{w}^T \mathbf{x}_i})) \end{aligned}$$

- ▶ Minimize: $\sum_{i=1}^N (-y_i \mathbf{w}^T \mathbf{x}_i + \log(1 + e^{\mathbf{w}^T \mathbf{x}_i}))$

LOGISTIC REGRESSION LEARNING

$$\text{minimize} \sum_{i=1}^N (-y_i \mathbf{w}^T \mathbf{x}_i + \log(1 + e^{\mathbf{w}^T \mathbf{x}_i}))$$

LOGISTIC REGRESSION LEARNING

$$\text{minimize } \sum_{i=1}^N (-y_i \mathbf{w}^T \mathbf{x}_i + \log(1 + e^{\mathbf{w}^T \mathbf{x}_i}))$$

$$\begin{aligned} \frac{d \log L(\mathbf{w} | D)}{d w_j} &= \sum_{i=1}^N \left(-y_i x_{ij} + \frac{1}{1 + e^{\mathbf{w}^T \mathbf{x}_i}} e^{\mathbf{w}^T \mathbf{x}_i} x_{ij} \right) \\ &= \sum_{i=1}^N \left(-y_i + \frac{1}{1 + e^{\mathbf{w}^T \mathbf{x}_i}} e^{\mathbf{w}^T \mathbf{x}_i} \right) x_{ij} \\ &= \sum_{i=1}^N (-y_i + P(y_i = 1 | \mathbf{w})) x_{ij} \end{aligned}$$

Convex!

But no closed form solution!

LR LEARNING

- Score function: likelihood function

$$\text{minimize } \sum_{i=1}^N (-y_i \mathbf{w}^T \mathbf{x}_i + \log(1 + e^{\mathbf{w}^T \mathbf{x}_i}))$$

- Estimate optimal $\mathbf{w}$ using gradient descent

Gradient descent:

Start at some $\mathbf{w}$, e.g., $\mathbf{w}=[0,0,0,0,0]$

Make predictions given current $\mathbf{w}$:

Calculate gradient for each parameter:

$$\begin{aligned} \forall i \quad \hat{y}_i &= P(y_i = 1 | \mathbf{x}_i) = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}_i}} \\ \forall j \quad \frac{d \log L}{d w_j} &= \left[\sum_{i=1}^n (-y_i + \hat{y}_i) x_{ij} \right] \\ &= \nabla_j \end{aligned}$$

Move parameters in direction of gradient: $\forall j \quad w_j^{new} = w_j - \eta \nabla_j$

Repeat

LR PREDICTION

Intercept	Age>40	Income=high	Student=yes	Credit=fair	BuysComp?
1	0	1	0	1	0
1	0	1	0	0	0
1	0	1	0	1	1
1	1	0	0	1	1
1	1	0	1	1	1
1	1	0	1	0	0
1	0	0	1	0	1
1	0	0	0	1	0
1	0	0	1	1	1
1	1	0	1	1	1
1	0	0	1	0	1
1	0	0	0	0	1
1	0	1	1	1	1
1	1	0	0	0	0
1	0	1	0	0	?

- What is the probability that new person will buy a computer?

$$\mathbf{x} = [1, 0, 1, 0, 0]$$

$$\mathbf{w} = [-.5, 1.2, 3, -2, 0.7]$$

$$\mathbf{x}^T \mathbf{w} = 0.7$$

$$P(BC = 1 | \mathbf{x}_i) = \frac{1}{1 + e^{-0.7}} = 0.668$$

DEAL WITH OVERFITTING

- ▶ Simply finding the parameter values that lead to maximum likelihood function value in the training dataset may imply overfitting!
- ▶ Solution: add a **regularization term** in the scoring function to penalize complex models
 - ▶ e.g., L2 regularization term: $\frac{\lambda}{2} \|w\|^2$
 - ▶ λ is the regularization parameter; the larger the value, the more we are in favor of simple models

LR LEARNING WITH REGULARIZATION TERM

- Score function: likelihood with L2 regularization

$$\text{minimize} \sum_{i=1}^N (-y_i \mathbf{w}^T \mathbf{x}_i + \log(1 + e^{\mathbf{w}^T \mathbf{x}_i})) + \frac{\lambda}{2} \|\mathbf{w}\|^2$$

- Estimate optimal $\mathbf{w}$ using gradient descent

Gradient descent:

Start at some $\mathbf{w}$, e.g., $\mathbf{w}=[0,0,0,0,0]$

Make predictions given current $\mathbf{w}$:

Calculate gradient for each parameter:

$$\begin{aligned} \forall i \quad \hat{y}_i &= P(y_i = 1 | \mathbf{x}_i) = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}_i}} \\ \forall j \quad \frac{d \log L}{d w_j} &= \left[\sum_{i=1}^n (-y_i + \hat{y}_i) x_{ij} \right] + \lambda w_j \\ &= \nabla_j \end{aligned}$$

Move parameters in direction of gradient: $\forall j \quad w_j^{new} = w_j - \eta \nabla_j$

Repeat

SVM: LEARNING

SVM: KNOWLEDGE REPRESENTATION AND SCORING FUNCTION

- ▶ Linear SVM: $y = \text{sign} \left[\sum_{i=1}^m w_i x_i + b \right]$
- ▶ Margin = $d_+ + d_- = 2/\|\mathbf{w}\|$
- ▶ Optimization problem
 - ▶ $\max 2/\|\mathbf{w}\|$
 - ▶ subject to
 $y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \forall i \in \{1, 2, \dots, N\}$

